

Exp 3 wireless technology

RFID and Keypad Password-Based Door Lock System

Aim:

To design and implement a secure door lock system using RFID and keypad-based authentication with Arduino Mega. This project enhances security by combining two-factor authentication (RFID and password).

Theory:

An RFID and Keypad Password-Based Door Lock System is a modern security solution that uses Radio Frequency Identification (RFID) and a keypad to control access to a secure area.

1. RFID (Radio Frequency Identification):

- o RFID is a wireless communication technology that uses radio waves to identify and authenticate objects or individuals.
- o An RFID card or tag contains a unique identification number stored in its microchip.
- o When the card is brought close to the RFID reader, the reader detects the ID and sends it to the Arduino for verification.

2. Keypad Authentication:

- o A keypad is used to input a numeric or alphanumeric password.
- o If the entered password matches the stored password, the system proceeds to unlock the door.
- o This adds an additional layer of security to the system.

3. Servo Motor and Lock Mechanism:

- o A servo motor is used to control the locking mechanism of the door.
- o When the correct card and password are detected, the servo rotates to unlock the door.
- o If the authentication fails, the door remains locked.

4. LCD Display:

- o The LCD display provides real-time feedback to the user.
 - o It shows messages like "Access Granted" or "Access Denied" based on the authentication result.
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Components:

- Arduino Mega — Acts as the central controller.
 - RFID Module (RC522) — Reads the RFID card/tag ID.
 - Keypad (4x4) — Takes user input for password authentication.
 - Servo Motor — Controls the door lock mechanism.
 - I2C LCD Display Module (16x2) — Displays system status.
 - Jumper Wires — For connecting components.
 - Breadboard — For easy wiring and component placement.
 - RFID Cards and Tags — Used for authentication.
 - Lock Mechanism — Physical locking system.
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Working:

1. **System Initialization:**
 - a The Arduino initializes and displays "System Ready" on the LCD.
 - a The servo motor sets the door to a locked position.
 2. Reading RFID Card:
 - a When the user places an RFID card near the reader, the reader reads the unique ID.
 - a If the ID matches the stored value, the system proceeds to the next step.
 3. Password Verification:
 - a The user is prompted to enter the password using the keypad.
 - a If the password is correct, the servo unlocks the door and displays "Access Granted."
 - o If the password is incorrect, "Access Denied" is shown on the LCD.
 4. Locking the Door:
 - a After a few seconds, the servo motor returns to the locked position.
 - a The system resets and waits for the next input.
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Code:

```
#include <Servo.h>
#include <Wire.h>
#include <LiquidCrystal I2C.h>

Servo myservo;
LiquidCrystal I2C lcd(0x27, 16, 2);
```

```
String password = "1234"; // Set the correct password
String input = "";
```

```
void setup() {
  lcd.begin();
  lcd.print("System Ready");
  myservo.attach(9);
  myservo.write(0); // Lock position

void loop() (
  char key = keypad.getKey();
  if (key) (
    input += key;
    lcd.setCursor(0, 1);
    lcd.print(input);

    if (input.length() == 4) (
      if (input == password) (
        lcd.clear();
        lcd.print("Access Granted");
        myservo.write(90); // Unlock position
        delay(3000);
        myservo.write(0); // Lock again
      } else {
        lcd .clear();
        lcd .print( "Access  Denied");
        delay( 1000);

        :tinput
        lcd .clear();
        lcd .pr:Int( "System  Ready" );
```

Conclusion:

The RFID and Keypad Password-Based Door Lock System is a secure and efficient solution for access control. By combining RFID technology with password authentication, it provides

two-layer security. The system is user-friendly, cost-effective, and suitable for various security applications.